

in annual health care expenditures each year [4]. As a result, improving medication adherence continues to be recognized as a critical priority in diabetes management.

Medication adherence is a multidimensional behavior that extends beyond simply taking medications as prescribed. According to the widely adopted ABC taxonomy, adherence comprises three interrelated phases: initiation (starting a prescribed medication), implementation (the extent to which medication-taking corresponds with the prescribed regimen), and persistence (the duration of continued treatment before discontinuation) [5]. These phases reflect distinct aspects of medication-taking behavior and may be influenced by different behavioral, clinical, and contextual factors. Medication adherence can be evaluated using a variety of approaches, including self-reported adherence questionnaires, pharmacy refill records, electronic monitoring systems, medication possession measures, and clinical indicators [6]. Because each method captures different dimensions of adherence and varies in its sensitivity to behavioral change, recognizing adherence as a multidimensional construct is essential for interpreting the heterogeneous interventions and outcomes reported in the literature [7].

Medication-taking behaviors in T2D are affected by multiple interacting determinants that extend beyond clinical factors alone. Patients' beliefs about treatment necessity, concerns about adverse effects, complexity of medication regimens, forgetfulness, and structural barriers related to medication access or health services can all influence adherence behaviors [8]. Challenges with medication adherence may arise from intentional decisions, such as when patients question the benefits of pharmacologic therapy or prefer to rely on lifestyle modification when the provider has recommended or prescribed a medication regimen [9]. In other situations, difficulties with medication-taking reflect unintentional barriers, including forgetfulness, competing daily priorities, or difficulties integrating medications into routine activities [10]. Because T2D requires lifelong pharmacologic management, effective adherence interventions must account for both the behavioral and contextual factors that influence medication-taking.

The factors influencing medication adherence extend beyond individual beliefs and behaviors to include broader social and structural conditions. Financial barriers, access to healthcare services, health literacy, housing instability, food insecurity, transportation challenges, work-related demands, caregiving responsibilities, and social support can all influence an individual's ability to initiate, implement, and sustain medication use [11]. These factors may also contribute to disparities in treatment engagement and diabetes outcomes. Their importance becomes even more apparent in the context of digital adherence interventions, where access to smartphones, reliable internet connectivity,

digital literacy, and ongoing engagement with healthcare services are often prerequisites for successful use [12]. As a result, social and structural conditions may influence not only medication adherence itself but also the reach, uptake, and effectiveness of digital health interventions. Recognizing these broader influences is important when interpreting intervention effectiveness and considering how digital adherence strategies can be implemented equitably across diverse populations.

In recent years, digital health technologies have received increasing attention as important tools to support medication adherence and chronic disease self-management [13]. Advances in mobile connectivity, wearable sensors, and telemedicine platforms have expanded opportunities to deliver behavioral support beyond traditional in-person care [14]. Mobile health interventions, including smartphone applications (apps), remote glucose monitoring systems, and telemedicine services, enable patients to receive education, reminders, and feedback in real time while allowing clinicians to monitor treatment progress remotely [15]. These technologies may reduce geographic and logistical barriers that limit access to health care services, particularly for individuals living in rural or medically underserved settings [16]. Digital platforms can also facilitate sustained engagement by integrating reminders, personalized feedback, and interactive communication between patients and healthcare professionals [17]. Because of their scalability and relatively low unit cost, digital interventions have generated considerable interest as potential strategies to improve treatment adherence and long-term disease outcomes [18]. A wide range of digital tools has been explored to support medication-taking behaviors in diabetes management [15, 19]. These include mobile apps that provide educational and medication reminders [20], telemedicine platforms that facilitate remote consultations [21], short message service (SMS) systems, automated phone calls, and web-based patient portals [22]. More recent innovations have incorporated additional features, such as support for insulin titration, video-assisted monitoring, and sensor technologies that help track medication use over time [23].

Although digital interventions for diabetes self-management have been evaluated in several prior reviews, most have focused on broad behavioral or clinical outcomes rather than medication adherence as a distinct therapeutic target [24–26]. Furthermore, the rapid evolution of digital health technologies since 2021, including remote patient monitoring, integrated digital platforms, conversational artificial intelligence (AI), and increasingly personalized approaches, has expanded both the range of available interventions and the ways in which they are delivered [14]. As these technologies continue to evolve, digital interventions have also become more complex, frequently combining medication reminders, educational content, self-monitoring, personalized feedback, and patient-provider communication within